

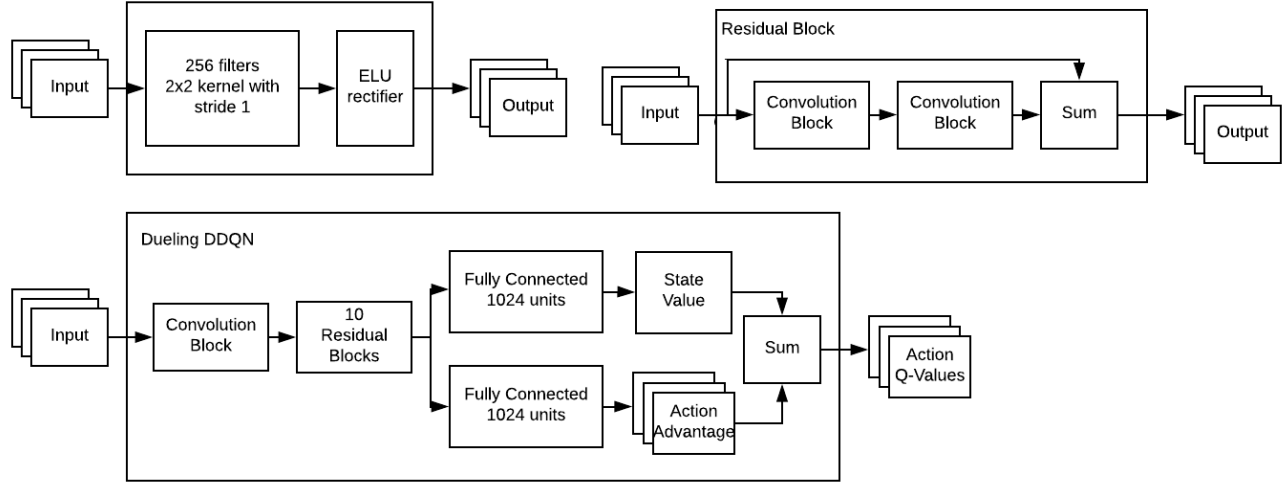


Figure 4: The architecture of our DDDQN consists of a convolutional layer followed by a tower of 10 residual blocks. The final residual block is fed to two separate fully-connected layers to produce the current states Q-value and the predicted advantage of each possible action. The streams are combined to produce predicted action Q-values.

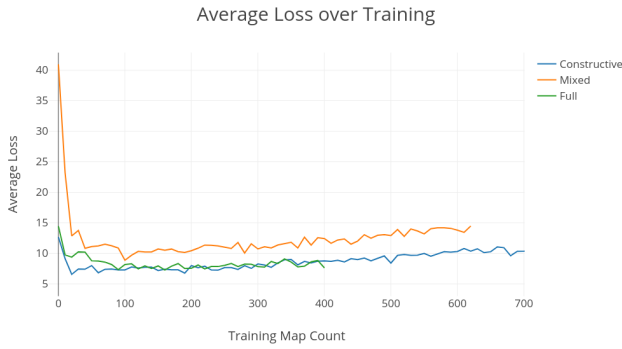


Figure 5: The loss over time for the training of each network. Loss was collected every 5 maps of training by averaging the loss across the 5 maps.

from previously evolved maps to prevent the network from learning to recognize evolved maps.

Given the generalizability of ECCL as in the discussion from the previous section, as it only requires a data generator and a game-playing agent architecture, we also expect it to work well with AlphaGo Zero-based agents as well, and leave that open for future work.

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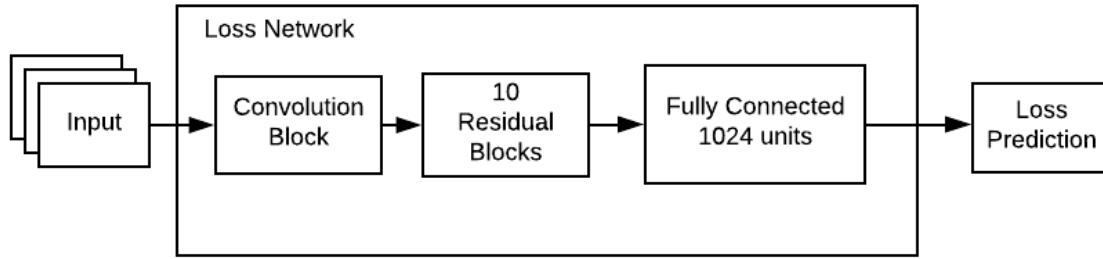


Figure 6: The architecture of our loss network consists of a convolutional layer followed by a tower of 10 residual blocks which feeds a fully connected layer that outputs the loss prediction.

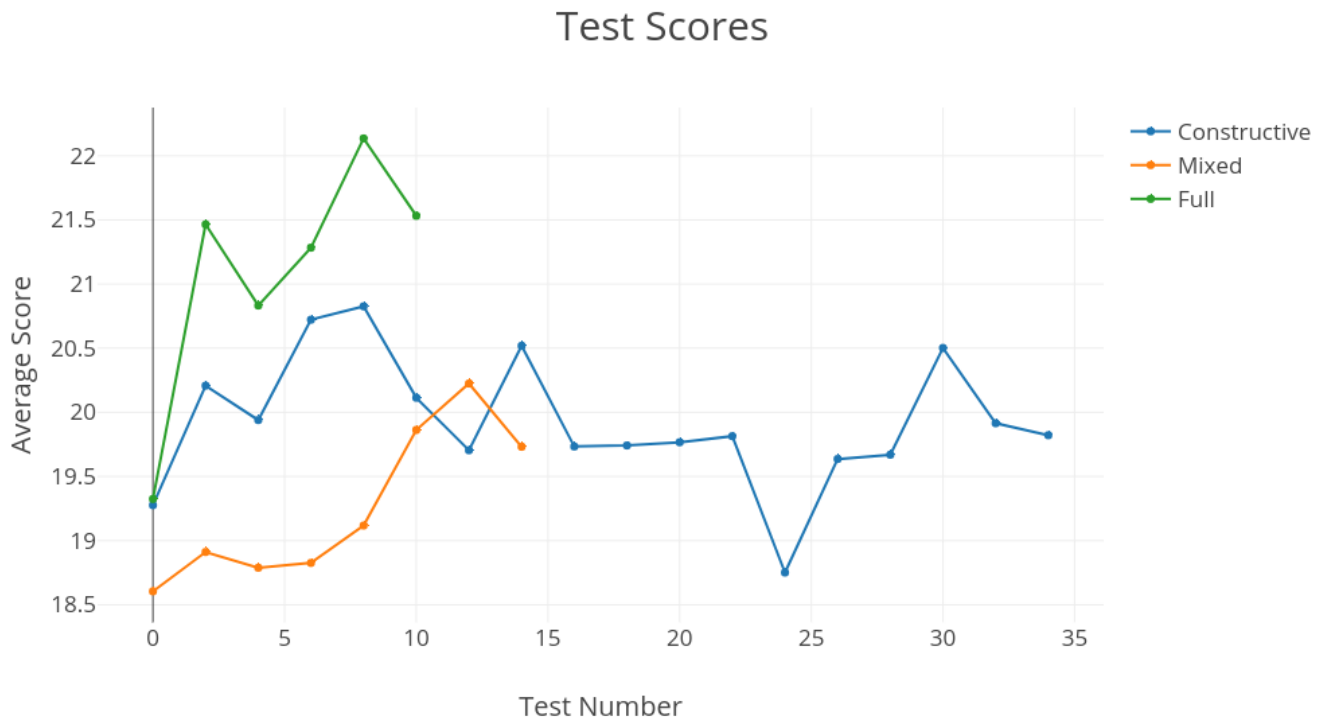


Figure 7: The testing score results averaged in 1,000 map batches over the course of training. The y-intercept marks network performance at network initialization with randomized weights.

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